

HENAN, China

WMO: 560650

Lat: 34.7343N

Lon: 101.6011E

Elev: 11486

StdP: 9.54

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-10.5	-25.1	2.1	-1.0	-21.4	2.6	3.3	24.4	31.0	21.7	28.4	0.7	90	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	68.8	53.2	66.0	51.5	63.4	49.9	55.6	65.6	54.0	62.9	52.5	60.4	6.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.0	89.2	60.9	50.6	84.7	58.8	49.3	80.6	57.2	29.2	65.8	27.9	63.2	26.7	60.4	61.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.3	16.8	14.2	-22.4	73.8	4.2	2.0	-25.4	75.3	-27.9	76.4	-30.2	77.6	-33.3	79.1		
			-22.9	58.7	4.1	1.3	-25.8	59.6	-28.2	60.4	-30.5	61.1	-33.5	62.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.8	11.6	17.5	26.5	36.3	42.5	48.3	52.2	51.7	45.8	35.4	23.4	13.6
	DBStd	15.31	6.77	7.25	6.25	5.50	4.57	4.50	4.31	4.73	5.45	5.55	6.42	5.91
	HDD50	6151	1190	909	729	412	235	85	25	36	150	452	799	1129
	HDD65	11384	1655	1329	1194	862	697	502	396	413	577	916	1249	1594
	CDD50	242	0	0	0	0	3	33	94	88	23	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.8	4.2	5.0	5.8	6.0	5.8	5.3	4.9	4.5	4.6	4.2	4.1	3.9
Precipitation	PrecAvg	23.5	0.2	0.2	0.5	1.1	2.8	3.8	5.0	4.3	3.8	1.5	0.2	0.1
	PrecMax	35.0	0.7	0.6	0.9	2.8	6.1	6.9	10.2	10.3	7.2	4.4	0.7	0.2
	PrecMin	17.8	0.0	0.0	0.0	0.3	1.6	1.6	2.0	1.7	1.8	0.3	0.0	0.0
	PrecStd	3.7	0.2	0.2	0.3	0.6	0.9	1.1	1.8	1.9	1.4	0.9	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	49.1	56.0	62.6	64.9	68.1	72.9	71.8	69.0	60.0	49.4	42.5
		MCWB	26.8	30.4	33.9	39.2	43.5	52.2	55.1	55.5	53.6	43.0	32.1	27.1
	2%	DB	36.8	42.7	49.5	57.1	60.8	64.1	69.0	68.4	64.0	54.2	44.7	37.3
		MCWB	23.8	27.2	31.3	37.3	42.6	48.9	53.4	53.5	50.3	39.9	29.8	24.3
	5%	DB	32.4	38.4	45.2	53.5	57.4	61.5	66.0	65.6	60.0	50.2	41.6	33.3
		MCWB	21.3	25.2	29.5	35.6	41.3	47.8	51.9	52.1	47.9	37.3	28.3	22.0
	10%	DB	28.1	34.3	41.2	49.7	54.2	58.7	63.0	62.6	56.5	46.8	38.1	29.5
		MCWB	18.9	23.1	27.6	34.5	40.2	46.8	50.9	50.7	46.0	35.9	26.7	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	30.6	35.4	42.5	48.4	55.7	57.6	57.7	56.0	46.8	33.2	27.8
		MCDB	41.3	47.3	51.9	56.7	58.5	64.7	67.7	68.7	65.5	55.3	46.4	41.4
	2%	WB	24.2	27.9	32.4	39.3	45.8	52.8	55.5	55.7	52.5	42.8	31.0	24.6
		MCDB	36.3	41.0	46.3	52.6	55.8	60.5	65.9	65.6	61.4	50.3	42.2	36.2
	5%	WB	21.6	25.7	30.4	37.4	43.7	50.4	54.0	54.0	50.2	39.8	29.3	22.3
		MCDB	32.0	37.4	42.7	49.2	52.9	57.8	63.1	62.8	57.7	47.0	39.3	32.7
	10%	WB	19.0	23.5	28.5	35.7	41.9	48.5	52.4	52.5	48.0	37.5	27.4	20.1
		MCDB	27.6	33.4	39.6	46.9	50.9	55.4	60.3	60.0	54.0	44.3	36.6	29.2
Mean Daily Temperature Range	5% DB	MDBR	33.4	30.9	26.1	22.7	20.9	18.2	20.0	20.5	20.1	21.8	28.7	33.3
		MCDBR	39.6	37.8	34.5	31.0	28.2	24.2	26.9	26.1	26.3	27.9	33.1	38.3
		MCWBR	27.6	24.1	18.4	13.5	12.4	11.1	12.7	12.4	13.1	14.5	19.3	26.5
	5% WB	MCDBR	37.8	35.0	29.6	24.8	21.2	18.5	20.9	20.3	21.0	21.4	29.3	36.5
		MCWBR	26.7	22.8	16.1	11.4	10.7	9.3	10.5	10.0	11.0	12.0	17.5	25.5
Clear-Sky Solar Irradiance	taub	0.220	0.256	0.308	0.343	0.340	0.331	0.322	0.317	0.292	0.265	0.229	0.207	
	taud	2.519	2.421	2.249	2.181	2.293	2.412	2.477	2.480	2.539	2.553	2.564	2.585	
	Ebn at Noon	324	318	306	298	299	300	302	303	309	312	317	325	
	Edh at Noon	25	31	41	46	42	37	35	34	30	27	24	22	
All-Sky Solar Radiation	RadAvg	1069	1319	1636	1872	1830	1748	1816	1710	1418	1282	1152	1030	
	RadStd	63	68	80	92	77	145	179	149	128	91	37	36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		+0.97	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)		+0.76	+1.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page